CS1910 Computer Science I • Repetitions **Partner workbook**

The Guessing Game and Nested Squares

In lecture with a partner • about 30 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

Two loop problems: adding a counter and an outer loop to a `while` loop you already have, and using computational thinking to plan a drawing made of loops inside loops.

1 The guessing game

about 15 minutes

Here is the guessing game from lecture:

```
import random
secret = random.randint(1, 10)   # a random number from 1 to 10, inclusive

guess = int(input("I'm thinking of a number between 1 and 10, guess it: "))
while guess != secret:
    guess = int(input("Nope! Guess again: "))

print("Yay you guessed correctly!")
```

1. **Warm-up:** change the program so that it counts how many guesses the user took, and prints the count at the end.

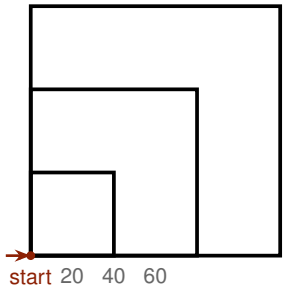
2. **Challenge:** after a correct guess, ask whether they want to play again. When they stop, print their **average** number of guesses over all the games.

3. If you played the game a thousand times, what would you expect the average number of guesses to be? (Hint: the game never says “higher” or “lower”.)

2 Nested squares

about 15 minutes

4. Use the turtle to draw this shape. Work through the four steps below *before* writing code.



5. **Decomposition:** what are the basic operations? What earlier work can you reuse?

6. **Patterns:** what are the patterns, and the patterns within patterns?

7. **Algorithm:** write your algorithm in pseudocode.

8. **Code:** now write it in Python.

✓ Before you leave

about 1 minutes

- [ ] I can add a counter and a running total to a loop.
- [ ] I can put one loop inside another and explain what each loop repeats.
- [ ] I can plan with decomposition, patterns and pseudocode before coding.